

Discovery Executive Summary

Project: test-discovery · Generated: 18/08/2026, 18:58:40

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 2 discovery analyses run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—
2	Frontend Modernization Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

The Klearcom platform is a monorepo with a Laravel 12 backend (6 API controllers, 4 Eloquent models, 2 services), a React 19 SPA frontend (4 page components, 4 shared components, 1 custom hook, 1 Zustand store), and a Node/Express dev-API (6 source files). Backend controllers are lean on average (74 LOC) but 4 of 6 access Eloquent models directly — there is no repository layer, and business logic (reachability calculations, tree building) is duplicated across controllers and services in 3 independent locations. The `LegacyReportController` crosses both Discovery and Connect domain boundaries, reading all four models, while `DashboardController` does the same for aggregation. The `LegacyDataMapper` uses PHP `extract()` — a dynamic-variable anti-pattern that creates implicit coupling. On the frontend, all 4 page components and 1 legacy class component make direct `api.*` calls with no service/data-access abstraction layer; the `LegacyMonitorPoller` is a class component with an interval memory leak. The two most urgent risks are (1) the missing repository pattern which scatters ORM access across 32 call sites, and (2) the cross-domain coupling in `LegacyReportController` and `DashboardController` that prevents independent evolution of Discovery and Connect modules. Layers covered: backend (26 PHP files), frontend (15 TS/TSX/JSX files), dev-api (6 JS files).

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	74 LOC avg (max 102)	Good
H2	Missing Service Layer	Controllers accessing repos/models	<15	15–30	>30	25 direct model calls in controllers	Moderate
H3	Missing Repository Pattern	Direct DB access points outside repos	<15	15–30	>30	32 (25 controllers + 7 services)	High Risk
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0	Good
H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	1 (LegacyDataMapper)	Moderate
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	100% (all Eloquent, no raw SQL)	Good
H7	God Classes	Largest class/file LOC	<500	500–600	>600	276 LOC (dev-api server.js); 222 LOC (ConnectPage.tsx); 165 LOC (MongoService.php)	Good
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	4 (LegacyReportController: 4 cross-domain models; DashboardController: 2)	Moderate
H9	Shared Database Coupling	Tables shared across domains	<30%	30–40%	>40%	~33% — 2 of 4 models queried by cross-domain controllers	Moderate
H10	Duplicated Business Logic (additional)	Independent copies of same algorithm	0	1–2	>2	3 copies of reachability formula + 3 copies of buildTree	Moderate

H11	Missing Frontend Service Layer (additional)	Page components with direct API calls	0	1–3	>3	5 components with 14 total direct api.* calls	Moderate
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1.4 Actions Required

Hotspot	Action	Rating	Priority
H3. Missing Repository Pattern	Introduce <code>ConnectMonitorRepository</code> and <code>DiscoveryJobRepository</code> interfaces with Eloquent implementations; bind in <code>AppServiceProvider</code> ; replace all 32 direct <code>Model::</code> calls	High Risk	High
H2. Missing Service Layer	Create <code>ConnectReachabilityService</code> , <code>DiscoveryTreeService</code> , <code>DashboardKpiService</code> ; move business logic out of 4 controllers	Moderate	High
H10. Duplicated Business Logic	Consolidate 3 copies of reachability formula and 3 copies of <code>buildTree()</code> into single canonical service methods	Moderate	High
H8. Domain Boundary Violations	Split <code>LegacyReportController</code> by domain; route cross-domain reads through service interfaces	Moderate	Medium
H9. Shared Database Coupling	Define table ownership per domain; expose read-only service APIs for cross-domain consumers	Moderate	Medium
H5. Shared Utility Abuse	Remove <code>extract()</code> from <code>LegacyDataMapper</code> and <code>LegacyReportController</code> ; move mapping logic to domain service	Moderate	Medium
H11. Missing Frontend Service Layer	Create <code>connectService.ts</code> / <code>discoveryService.ts</code> ; extract domain hooks; convert <code>LegacyMonitorPoller</code> to functional component with cleanup	Moderate	Medium

1.5 Expected Outcomes

- Testability:** Repository interfaces enable unit testing of services without a database; frontend service layer enables mocking API calls in component tests.
 - Single source of truth:** Consolidating the reachability formula and `buildTree()` into canonical service methods eliminates the risk of divergent business logic across endpoints.
 - Independent module evolution:** Enforcing domain boundaries (Discovery vs. Connect) through service interfaces means schema changes in one domain cannot silently break the other.
 - Onboarding velocity:** New contributors can work on Discovery or Connect independently without needing to understand both domains — bounded contexts reduce the knowledge surface area.
 - Migration readiness:** Repository abstraction + domain services position the codebase for future extraction into separate deployable services, should the platform scale beyond a monolith.
- ```

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2. Frontend Modernization Analysis

EXECUTIVE SUMMARY

The Klearcom frontend is a compact React 19 + TypeScript single-page application consisting of 13 source files and 11 component definitions, built with a modern Vite toolchain and well-chosen library stack (Zustand for client state, React Query for server cache). The most severe gap is **significant structural and code duplication** between the two primary feature pages — `ConnectPage` and `DiscoveryPage` — which share near-identical form layouts, query patterns, invalidation logic, and MongoDB transcript rendering blocks that should be extracted into shared components and hooks. A legacy class-based component (`LegacyMonitorPoller.jsx`) with an **uncleared interval leak** and a `LegacyDashboardWidget` that **throws unhandled errors without an Error Boundary** represent resource safety and crash-resilience risks. Component sizes are well within healthy limits and state management is clean and scoped.

3.1 Benchmark Ratings Summary

| #  | Hotspot                       | Primary KPI                 | <span class="rating rating-good">Good | <span class="rating rating-moderate">Moderate | <span class="rating rating-high-risk">High Risk | Measured                      | Rating                                          |
|----|-------------------------------|-----------------------------|---------------------------------------|-----------------------------------------------|-------------------------------------------------|-------------------------------|-------------------------------------------------|
| H1 | UI Component Duplication      | Duplicate components %      | <5%                                   | 5–10%                                         | >10%                                            | ~33% (3 of 9 component files) | <span class="rating rating-high-risk">High Risk |
| H2 | Legacy Class-Based Components | Modern component adoption % | >90%                                  | 70–90%                                        | <70%                                            | 91% (10 of 11 definitions)    | <span class="rating rating-good">Good           |

|    |                                       |                                                                    |      |         |      |                                   |                                                 |
|----|---------------------------------------|--------------------------------------------------------------------|------|---------|------|-----------------------------------|-------------------------------------------------|
| H3 | Massive Components                    | Largest component LOC                                              | <400 | 400–500 | >500 | 222 LOC (ConnectPage.tsx)         | <span class="rating rating-good">Good           |
| H4 | Global State Dependencies             | Components reading global state %                                  | <30% | 30–60%  | >60% | 18% (2 of 11 components)          | <span class="rating rating-good">Good           |
| H5 | Complex State Management              | Max prop-drilling depth                                            | <3   | 3–5     | >5   | 1–2 levels                        | <span class="rating rating-good">Good           |
| H6 | Missing Error Boundaries (additional) | Error boundary coverage (%; target 100% of throw-capable subtrees) | >80% | 50–80%  | <50% | 0% (no ErrorBoundary in codebase) | <span class="rating rating-high-risk">High Risk |
| H7 | Resource / Memory Leak (additional)   | Components with uncleared timers/subscriptions (count; target 0)   | 0    | 1       | >1   | 1 (LegacyMonitorPoller)           | <span class="rating rating-moderate">Moderate   |

3.4 Actions Required

| Hotspot                       | Action                                                                                                                                                                      | Rating                                          | Priority                                |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------|
| H1 — UI Component Duplication | Extract shared <code>TranscriptList</code> component, create <code>useModulePage</code> hook to deduplicate page structure, delete <code>LegacyDashboardWidget.tsx</code>   | <span class="rating rating-high-risk">High Risk | <span class="sev sev-critical">Critical |
| H6 — Missing Error Boundaries | Add <code>ErrorBoundary</code> at app root and per-route level; replace <code>throw new Error</code> with rendered error state                                              | <span class="rating rating-high-risk">High Risk | <span class="sev sev-high">High         |
| H7 — Resource / Memory Leak   | Convert <code>LegacyMonitorPoller</code> to functional component with <code>useQuery</code> <code>refetchInterval</code> , or add <code>componentWillUnmount</code> cleanup | <span class="rating rating-moderate">Moderate   | <span class="sev sev-medium">Medium     |

3.5 Expected Outcomes

- Shared component library reduces duplication:** Extracting `TranscriptList` and `ModulePageLayout` eliminates ~33% structural duplication and makes adding new feature modules a configuration exercise rather than a copy-paste operation.
- Error Boundaries improve resilience:** Per-route boundaries ensure a failing API in one module cannot crash the entire platform — critical for a voice monitoring dashboard that must stay available.
- Functional component conversion improves safety:** Replacing the class-based `LegacyMonitorPoller` with a `useQuery`-based hook eliminates the interval memory leak and aligns with the codebase's established React Query patterns.
- Deletion of legacy widget reduces maintenance surface:** Removing `LegacyDashboardWidget` eliminates redundant network calls and a throw-without-boundary crash vector.
- Consistent patterns accelerate onboarding:** A single page layout pattern with hooks-based data fetching gives new contributors one clear way to build feature pages.  
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